PromptHSI: Universal Hyperspectral Image Restoration With Vision–Language Modulated Frequency Adaptation

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***Abstract*—Recent advances in all-in-one (AiO) RGB image restoration have demonstrated the effectiveness of prompt learning in handling multiple degradations within a single model. However, adapting these approaches to hyperspectral image (HSI) restoration poses significant challenges due to the domain gap between RGB and HSI features, information loss in visual prompts under severe composite degradations, and difficulties in capturing HSI-specific degradation patterns via text prompts. To address these limitations, we introduce PromptHSI, the first universal AiO HSI restoration framework that is both prompt-guided and frequency-aware. By incorporating frequency-aware feature modulation, which uses frequency analysis to decouple complex composite degradations, and by employing vision–language model (VLM)-guided prompt learning, our approach decomposes text prompts into intensity and bias controllers that effectively guide the restoration process while mitigating domain discrepancies. Extensive experiments demonstrate that our unified architecture excels at both fine-grained detail recovery and global information restoration across diverse degradation scenarios, highlighting its significant potential for practical remote sensing applications.**

***Index Terms*—All-in-one (AiO) image restoration, frequency modulation, hyperspectral image (HSI) restoration, vision–language model.**

I. Introduction

HYPERSPECTRAL images (HSIs) contain far more spectral bands than RGB images, providing much richer spectral detail [57]. Because different materials exhibit distinct reflectance patterns across wavelengths, each substance has a characteristic spectral signature that enables material identification [1], [17], [47], [57]. HSIs can capture this information and therefore support a wide range of high-value applications, such as land-cover classification and agricultural monitoring [33], [40]. However, in practical airborne and low-Earth-orbit (LEO) satellite sensing, HSIs are often degraded by a *mixture* of factors [17], including sensor noise, stripe noise, spatial blur, cloud/haze occlusion, and even band-level spectral loss caused by data transmission issues, calibration errors, or atmospheric conditions. These degradations are not only multisource and scene dependent but also induce spectral variability, where the spectral signatures of the same material may shift or become locally distorted across bands. Consequently, HSI restoration becomes a crucial preprocessing step for recovering degraded pixels from imperfect observations before downstream applications.

In summary, existing HSI restoration approaches fall into two lines: 1) model-based low-rank/sparse formulations that assume simple or single-type degradations and 2) promptable all-in-one (AiO) RGB restorers that lack spectral awareness and operate mainly in the spatial domain. The former cannot cope with mixed and band-dependent degradations, while the latter cannot accurately align textual descriptions with hyperspectral frequency patterns. This gap motivates a restoration framework that is both prompt-guided and frequency-aware.

To address these issues, low-rank tensor models [43], [53] are extensively used in HSI restoration, relying on the assumption that the clean image $\mathbf{X}$ exhibits low-rank properties [13], [55] across the spatial and spectral dimensions. The restoration process is typically formulated as $\mathbf{X} \approx \mathbf{L} + \mathbf{S} + \mathbf{N}$, where $\mathbf{L}$ represents the underlying low-rank structure of the clean image, $\mathbf{S}$ represents structured sparse noise (e.g., stripe noise or dead pixels), and $\mathbf{N}$ represents random noise, such as Gaussian or Poisson noise. However, this assumption typically fails under multiple degradations. The structured noise components $\mathbf{S}$ introduces high-rank artifacts, thereby increasing the observed rank: $\text{rank}(\mathbf{X}) \gg \text{rank}(\mathbf{L})$, which makes low-rank approximations less accurate. Furthermore, mixed noise complicates the separation between the clean component $\mathbf{L}$ and noise components $\mathbf{S}$ and $\mathbf{N}$, typically resulting in oversmoothing

$$\min_{\mathbf{L}} \|\mathcal{P}_{\Omega}(\mathbf{X} - \mathbf{L})\|_F^2 + \lambda \text{TV}(\mathbf{L}) \tag{1}$$

where P_{Ω} denotes a projection operator on the observed pixels, and TV($\mathbf{L}$) denotes the total variation regularization

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term. Although this enforces spatial smoothness, it typically suppresses fine details, degrading spectral fidelity. Consequently, when multiple degradations are present, the fundamental low-rank assumption is violated, rendering conventional tensor-based approaches ineffective. More adaptive methods that incorporate hybrid priors [36], [37], [73] or deep learning (DL)-based corrections are required to achieve robust restoration.

At the same time, AiO RGB restoration methods [4], [9], [10], [15], [38], [48] have shown that promptable networks can unify deblurring, deraining, dehazing, and denoising in a single model. They achieve this by injecting visual or textual prompts to modulate intermediate features. However, directly porting these AiO designs to HSIs is nontrivial. First, HSIs exhibit much higher spectral dimensionality and interband correlations than RGB, where the same semantic prompt (e.g., "cloudy" or "striping") can correspond to very different frequency patterns across bands. Second, existing AiO models are mostly designed in the spatial domain, whereas many HSI degradations (cloudy, band-missing) are more naturally characterized in the frequency domain.

To address these challenges, we propose **PromptHSI**, a frequency-aware, vision–language modulated HSI restoration framework. Unlike conventional HSI restoration approaches, PromptHSI lets a vision–language model (VLM) parse a textual description of the degradation and use it to steer a frequency-domain adapter that operates on HSI features. In this way, the model can adjust which frequency bands to suppress or recover according to the described corruption, instead of relying on a fixed, task-specific prior, enabling robust and adaptive restoration across diverse degradation scenarios. Our key contributions are summarized as follows.

1) *Challenge-Driven formulation for HSI Restoration:* We explicitly target HSI degradations that are mixed, band-dependent, and spectrally variable, and we show that low-rank or single-degradation assumptions are insufficient in such cases.
2) *VLM-Guided Frequency Adaptation:* We design a prompt-conditioned frequency module in which textual embeddings modulate the restoration behavior in the frequency domain. This enables controllable and interpretable restoration and distinguishes our method from prior AiO RGB restorers that modulate only spatial features.
3) *Composite Degradation Dataset for Benchmarking HSI Restoration:* To facilitate HSI restoration research, we construct a standardized composite-degradation dataset that incorporates cloud occlusion, blurring, noise, and spectral band loss. We evaluate PromptHSI against strong recent baselines and observe consistent gains under both perfect and imperfect prompts. The dataset is released to support future developments in HSI restoration.
4) *Real-World Validation and Prompt-Aware Separability:* We validate PromptHSI on real degraded HSIs without paired ground truth, using no-reference quality metrics and expert visual assessment, demonstrating practical feasibility and strong generalization. Moreover, quantitative cluster separability (Silhouette [52] and normalized mutual information (NMI) [46]) and t-SNE analyses show that prompts steer the network to form well-separated, degradation-aware representations, indicating that the model correctly infers the degradation type from the provided prompt.

Through extensive experiments, we demonstrate that PromptHSI consistently outperforms state-of-the-art AiO RGB methods in HSI restoration tasks while providing controllable restoration via text prompts (see Fig. 1).

II. Related Works

A. HSI Restoration

HSI restoration is a fundamental task in remote sensing because real-world hyperspectral data are often incomplete or degraded. Early works mainly relied on optimization-based methods [68], and we refer to these as *conventional* approaches. More recently, DL-based methods have driven a new wave of research on HSI restoration. Below, we briefly review these two branches of work.

1) Conventional Approaches: Before the advent of DL, most HSI restoration methods were built on structural properties of HSIs, such as low-rankness, sparsity, and nonlocal spatial–spectral similarity. Qian and Ye [49] employed sparse representation together with nonlocal spectral–spatial structures for denoising. FastHy [73] further incorporated low-rank priors to accelerate hyperspectral denoising. RhyDe [74] adopted a collaborative sparsity formulation to better preserve rare pixels for both denoising and anomaly detection. WSRRR [51] introduced wavelet-based priors and a cyclic-descent optimization scheme. Jia et al. [23] remove stripe noise using statistical information. Other representative methods [5], [45], [53] exploited 3-D total variation and handcrafted priors to separate clean signals from noise and stripe artifacts, and some of them can even be extended to HSI super-resolution.

In parallel, spectral variability has been extensively studied in the hyperspectral unmixing literature. For example, the augmented linear mixing model (ALMM) [18] explicitly introduces variability terms to account for illumination changes and material-dependent variations at the endmember level, thereby improving material identification under varying conditions. Such unmixing-oriented models are complementary to our restoration-stage framework: PromptHSI focuses on recovering cleaner, frequency-consistent HSI cubes that can subsequently be exploited by spectral unmixing methods.

2) DL-Based Approaches: More recently, DL-based methods have sparked a new wave of research on HSI restoration. By learning end-to-end spatial–spectral mappings, these approaches can recover HSIs more effectively under complex noise patterns and diverse sensing conditions than purely model-based methods [2], [6].

Classic DL-based HSI restoration networks, such as HSIDeNet [6] and En-OCTNet [58], use CNNs to remove Gaussian noise and structural stripes. Subsequent works further improved the network architecture to better exploit the joint spatial–spectral structure of HSIs. For example, SST [31] introduces transformers along both the spectral and spatial dimensions to capture spectral–spatial correlations, while

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SERT [32] designs a multishape rectangular self-attention module to more comprehensively explore nonlocal spatial similarity. Other approaches replace standard 2-D convolutions with 3-D convolutions, sometimes combined with self-attention or quasi-recurrent pooling, to extract volumetric HSI features more effectively [54], [62]. There are also recent task-specific restoration models targeting spatial blurring [21], [24], [34], limited spectral resolution [12], [16], [27], or even full spectral reconstruction [3], [29].

In addition, several methods integrate HSI-specific priors into DL frameworks. ADMM-ADAM [37] embeds a convex optimization perspective into a learned architecture to obtain a simple and effective regularization form, and Sidorov and Hardeberg [55] exploit the intrinsic properties of untrained CNNs to perform various restoration tasks, thereby alleviating the heavy data requirements of fully supervised DL-based methods.

Despite these advances, most of the above methods are designed and trained for a single, well-defined degradation (e.g., noise, stripe, spatial blur, cloud/haze occlusion, and band-level spectral loss). These methods typically train models separately for each type of degradation. However, in real remote-sensing scenarios, HSIs can suffer from multiple degradations at once (e.g., noise + striping + cloud occlusion). Under such composite conditions, models that have only seen isolated degradations are easily misled, leading to what we may call erroneous compensation: the network tries to denoise an occluded region or to destripe a band-missing area, which amplifies artifacts and disrupts spectral consistency [28], [39]. For instance, when both sensor noise and cloud occlusion are present, a denoising-oriented model may incorrectly regard clouds as random noise and oversmooth them, resulting in spectral distortions and suboptimal restoration outcomes.

B. AiO RGB Image Restoration

Compared with HSI restoration, which must jointly model global context and local spatial–spectral correlations [59], [67], RGB image restoration is typically concerned only with spatial structures. This is largely because RGB images have only three channels and the degradations considered in RGB benchmarks are generally less diverse and less coupled than those observed in practical hyperspectral sensing [17], [30].

Building on the simpler setting, a recent line of work in RGB restoration has explored AiO models that handle multiple degradations within a single network [4], [7], [9], [14], [15], [28], [48]. These multitask frameworks leverage prompts or task-aware modules to share features across degradations. For example, AirNet [28] uses contrastive learning to jointly learn degradation representations and restoration; PromptIR [48] adopts visual prompts for degradation perception during decoding; InstructIR [9] employs text prompts to guide the network; and HAIR [4] generates degradation-specific parameters via a hypernetwork. OneRestore [15] further extends AiO restoration to composite corruptions, while AdaIR [10] transforms feature maps into the frequency domain and applies an adaptive mask to decouple low- and high-frequency components.

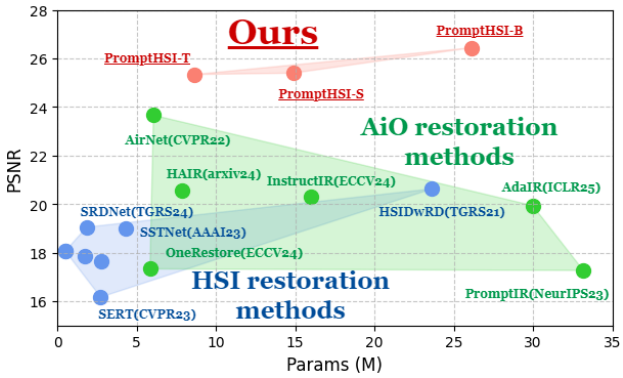


Fig. 1. Performance comparison between AiO and HSI restoration methods on the AVIRIS dataset. The proposed **PromptHSI** offers leading performance and controllability while retaining a compact model size.

However, despite their success in the RGB domain, these AiO strategies do not transfer directly to HSIs. The domain gap, the much higher spectral dimensionality, and the frequent presence of composite and band-dependent degradations make it difficult to learn compact degradation representations and to restore HSIs effectively using RGB-oriented designs.

III. Preliminary and Motivation

A. Hybrid Architecture for HSI Restoration

Effective HSI restoration requires an architecture capable of handling diverse degradations within a unified framework [8], [63]. Existing architectures primarily fall into two categories: 1) U-shaped encoder–decoder networks [60], [61], which effectively capture global semantic representations for overall restoration but experience difficulty with fine details; or 2) plain residual-in-residual networks [20], [35], [72], which enhance spatial features and high-frequency information but lack global awareness.

AiO restoration models [4], [15], [48] and HSI-specific methods [22], [25], [41] favor the former approach, limiting fine-grained detail recovery. To overcome this challenge, PromptHSI integrates residual-in-residual blocks within a U-shaped cascade decoder, preserving the global context while refining local details for greater adaptability across degradation scenarios.

B. Degradation Analysis in Frequency Domain

Unlike RGB image restoration, HSI restoration must account for complex spatial–spectral correlations across hundreds of spectral bands [67]. The degradation process can be mathematically formulated as $I_{\text{degraded}} = \mathcal{D}(I_{\text{clean}}) + \epsilon$, where $\mathcal{D}(\cdot)$ denotes the degradation operator, encompassing effects such as blurring, stripe noise, and occlusion, and ϵ denotes additive noise.

1) Frequency Modulation: Recovering Iclean from I_{degraded} using an arbitrary inverse degradation function $\mathcal{D}^{-1}\{\cdot\}$ is inherently challenging, particularly under composite degradations, where multiple degradation factors (e.g., noise, blur, and missing bands) interact in a complex manner. Unlike isolated degradations, which can typically be addressed using

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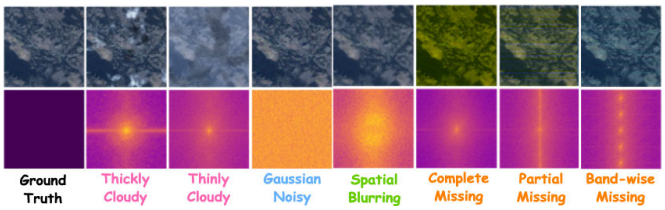


Fig. 2. Frequency-domain characteristics of HSI degradations. Fourier magnitude spectra visualization for different degradation types. (Top row) Fourier transforms of degraded HSIs. (Bottom row) Fourier transforms of residual images (degraded minus ground truth). Each degradation type manifests distinct frequency patterns—cloud occlusion affects low frequencies, Gaussian noise distributes across all frequencies, blurring attenuates high frequencies, and band-missing creates structured artifacts. This degradation-specific frequency behavior motivates our PGFM design.

predefined priors, composite degradations induce coupled distortions across both spatial and spectral dimensions, making conventional inverse modeling impractical.

Our empirical analysis (Fig. 2) reveals that different degradation types exhibit distinct but overlapping spectral characteristics. Motivated by this, we reformulate the following equation by modeling the degradation process as an affine transformation in the frequency domain:

$$F\left(I_{\text{degraded}}\right) = \lambda \odot F\left(I_{\text{clean}}\right) + \mu \tag{2}$$

where $F(\cdot)$ denotes the Fourier transform, λ modulates frequency amplitudes, μ shifts the frequency distribution, and $\odot$ represents element-wise multiplication. Under composite degradations, these frequency parameters exhibit the following complex dependencies.

1) Blurring attenuates high-frequency components ($\lambda_{\text{blur}} \ll \mu_{\text{blur}}$), reducing fine details.
2) Noise induces broadband perturbations, where ($\mu_{\text{noise}} \approx$ constant) remains nearly uniform across frequencies.
3) Occlusion and missing bands cause structured disruptions in both λ and μ, resulting in spectral inconsistencies.
4) Composite degradations result in frequency interactions, where overlapping effects create nontrivial distortions that extend beyond individual degradations.

For a clearer understanding, Fig. 3 (left) visualizes these degradation-specific frequency characteristics, highlighting the nonadditive nature of composite degradations in the Fourier domain.

2) Prompt Guidance: The diversity of HSI degradations presents significant challenges in learning the corresponding frequency coefficients. As shown in Fig. 3 (right), previous AiO restoration methods [4], [10] use adaptive modules to classify degradation types. However, lacking explicit prompt-based guidance, these methods solely rely on implicit feature learning, making parameter selection unreliable, particularly under composite degradations, where multiple degradation effects exhibit complex spectral interactions.

To address this, we introduce a VLM-driven prompt guidance mechanism, a core component of PromptHSI, that embeds frequency-aware priors in the restoration. Leveraging textual cues from a given prompt T, our approach modulates frequency coefficients, refining the parameter space for faster convergence and enhanced frequency adaptation. This ensures

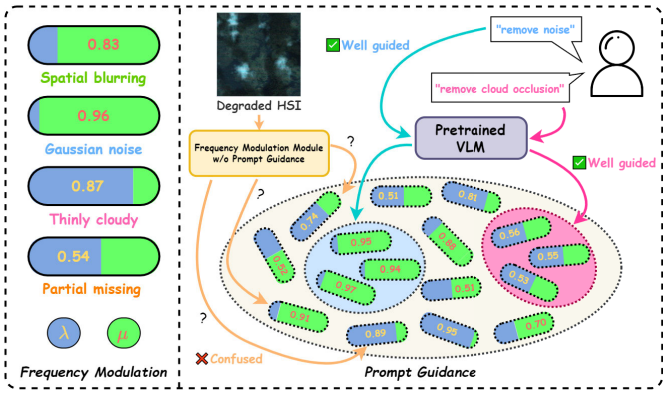


Fig. 3. Schematic of proposed strategy, where integrating prompt guidance with frequency modulation facilitates the adaptive selection of λ and μ, thereby enhancing HSI restoration under various degradation types. This mechanism encourages the model to efficiently converge toward optimal parameters.

precise control, adaptability, and robust performance across diverse degradations.

IV. Proposed Method

Overview: Fig. 4 illustrates the overall architecture of PromptHSI, which comprises the following three key components designed to address the challenges in universal HSI restoration.

1) A unified U-shaped network that contains a spatial–spectral aware feature extraction block (SSAEFB) and a prompt-guided feature aggregate block (PGFAB). This network effectively captures both enhanced spatial–spectral fine-grained details and global contextual information from degraded HSI.
2) A prompt-guided feature modulation (PGFM) that effectively bridges the domain gap between textual features and HSI features via frequency-domain modulation, while avoiding direct cross-domain feature alignment and information loss. Further details about the architecture of PromptHSI are provided in the Supplementary Material.

The primary contributions of this study to the literature are as follows.

1) *AiO RGB Paradigm:* Prior AiO approaches in RGB restoration typically realize controllability via *feature-* or *token-level* modulation (e.g., prompt- or instruction-driven gating, attention, or hypernetwork-generated parameters). Such designs operate in the RGB feature space and assume three-channel statistics, with no explicit spectral prior to guide frequency behavior under composite degradations.
2) *HSI-Specific Paradigm in PromptHSI:* In contrast, *PromptHSI* employs *frequency-aware controllers* that map text prompts to the intensity and bias parameters (λ, μ), which directly modulate the low-/high-frequency components of spatial–spectral features. This mechanism encodes physically meaningful HSI degradations (e.g., blur, noise, and band-missing) as frequency-domain interactions, thereby narrowing the restoration search space and improving spectral fidelity without relying on cross-domain feature alignment.

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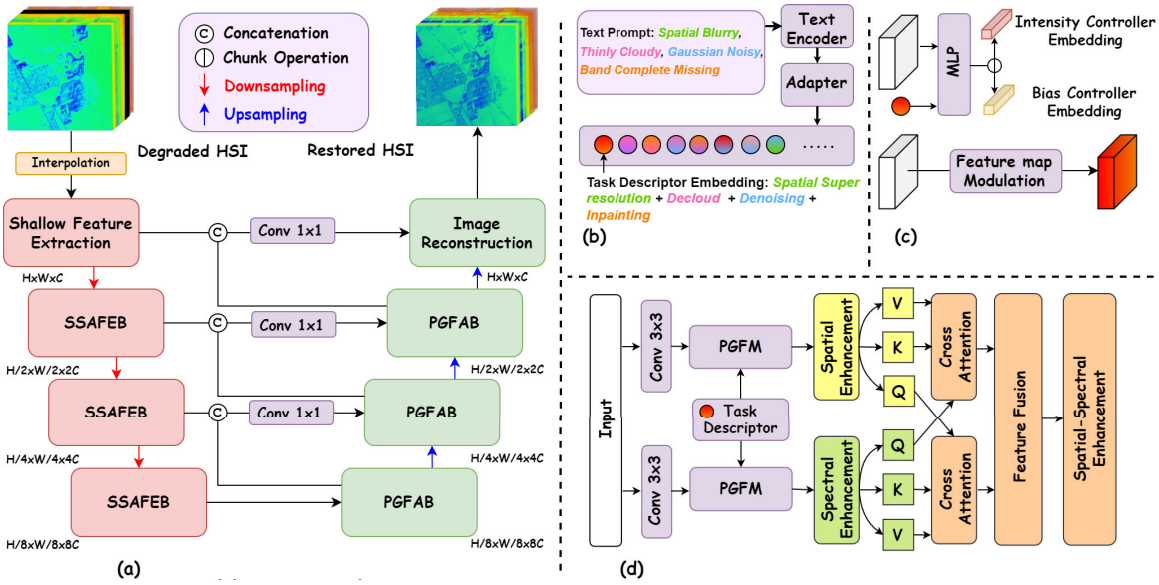


Fig. 4. Architecture of the proposed PromptHSI, which is designed to tackle the composite degradations within a single universal model via VLM-guided frequency modulation. First, we design a network based on a U-net-like encoder–decoder architecture for capturing global spatial–spectral information, while combining plain-residual-in residual networks in the proposed PGFAB for spatial–spectral enhancement. Afterward, text-prompt-guidance and frequency-aware feature modulation are proposed to jointly better capture the representation of composite degradation with controllability. (a) Overall pipeline of PromptHSI. (b) Task descriptor generation (TDG). (c) PGFM. (d) Prompt-guided feature aggregation block (PGFAB).

A. HSI Restoration Pipeline

Our restoration pipeline comprises two key components: an SSAEFB for efficient feature extraction and a PGFAB for controllable restoration with text-prompt guidance.

The encoder exploits the low-rank and sparse priors of HSIs through depth-wise separable convolution, effectively reducing computational complexity while preserving spectral characteristics. The PGFAB then uses dual branches for spatial and spectral enhancement, with cross-attention mechanisms facilitating comprehensive feature integration.

1) Spatial–Spectral-Aware Encoder: In conventional RGB restoration and other vision tasks, standard 3×3 convolutions are widely considered sufficiently versatile and effective feature encoders. However, considering the low-rank characteristics of HSI, where substantial redundancy exists between different spectral bands, using standard convolutions may incorporate unnecessary information from distant spectral bands, resulting in suboptimal performance. Therefore, encoders based on lightweight convolutions prove sufficiently effective while further reducing the parameter size to meet practical application requirements.

Specifically, for the input HSI I_{degraded}, we first interpolate the spectrally discontinuous regions (caused by the stripe effect) to facilitate the learning process. Subsequently, we apply standard convolution layers to obtain shallow feature maps.

Afterward, SSAFEB is based on a depth-wise separable convolution to extract features while inherently taking advantage of this low-rank prior, thereby reducing computational complexity without compromising performance [19].

2) Prompt-Guided Feature Aggregation Block: The features captured by the SSAFEB are combined with the informative text-prompt-based guidance from PGFM (which will be discussed in Section IV-B). Leveraging this semantic guidance, the network dynamically activates the corresponding weight for HSI restoration according to different degradation types. To capture detailed information, PGFAB utilizes two distinct branches: one focuses on the spatial dimension and the other on the spectral dimension. These branches are eventually integrated for feature aggregation.

3) Spatial and Spectral Feature Enhancement: The PGFM output is first refined using sequential RDG [20] and FRDB [19] modules to enhance fine details. Next, a cross-attention layer integrates spatial and spectral features from different branches. Specifically, we compute

$$\{Q_\alpha, K_\alpha, V_\alpha\} = \mathcal{F}^{\alpha}_{qkv}(F_\alpha), \quad \{Q_\beta, K_\beta, V_\beta\} = \mathcal{F}^{\beta}_{qkv}(F_\beta) \tag{3}$$

where F_α and F_β denote the spatial and spectral features, respectively. We then exchange queries between the branches to perform cross-attention as follows:

$$F^{\alpha}_{f} = \text{softmax}\left(\frac{Q_\beta K^{\top}_{\alpha}}{\sqrt{d_k}}\right) V_\alpha, \quad F^{\beta}_{f} = \text{softmax}\left(\frac{Q_\alpha K^{\top}_{\beta}}{\sqrt{d_k}}\right) V_\beta \tag{4}$$

where d_k denotes the scaling factor. The outputs F^{α}_{f} and F^{β}_{f} are fused and subsequently enhanced via a self-attention mechanism applied in both the spectral and spatial domains as follows:

$$\hat{F}^{\alpha\beta}_{f} = \mathcal{A}^{\alpha}\left(\mathcal{A}^{\beta}\left(\text{Conv}\left(\left[F^{\alpha}_{f}, F^{\beta}_{f}\right]\right)\right)\right) \tag{5}$$

where $\mathcal{A}^{\alpha}(\cdot)$ and $\mathcal{A}^{\beta}(\cdot)$ denote the self-attention enhancement blocks in the spatial and spectral domains, respectively, and $[\cdot]$ denotes the concatenation operator.

Our dual-branch architecture effectively integrates text-prompt-based guidance across the spatial and spectral dimensions. By embedding plain residual-in-residual modules [19], [20], [35] within a U-shaped cascade decoder [8], [61], the

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framework simultaneously captures local details and long-range dependencies, thereby achieving both fine-grained and global restoration to address diverse HSI degradations.

B. Prompt-Guided Feature Modulation

Directly using text features from VLMs for HSI restoration faces several critical challenges. First, VLM features are trained on RGB-text pairs, lacking an understanding of spectral characteristics and capturing complex spectral degradations. Second, their statistical distributions are mismatched because VLM features follow RGB image statistics, whereas HSIs have different data distributions, and direct adaptation requires complex domain translation.

Instead of directly bridging the HSI-text domain gap, the proposed PGFM addresses this challenge by decomposing the problem into two easier subtasks: 1) frequency-pattern matching with a task descriptor and 2) distribution adjustment by using a novel frequency-aware feature modulation mechanism. This strategy aims to selectively enhance the spectral fidelity across frequency bands, thereby overcoming the limitations of text-prompt-based guidance by avoiding direct cross-domain alignment between text features extracted from VLMs pretrained on RGB-text domains and HSI. Rather than performing cross-domain feature alignment, PGFM maps the CLIP text embedding to frequency-aware intensity (λ) and bias (μ) controllers that modulate the low- and high-frequency components of HSI features [see (9)], thereby avoiding the RGB–HSI statistics mismatch while preserving spectral fidelity.

1) Task Descriptor Generation: Given a text prompt T_{text} that provides the corresponding degradation description, we guide the HSI restoration network to obtain the specified restoration result. In this study, we employ CLIP [50], which integrates vision and language information and performs effectively in text feature extraction. We freeze the pretrained CLIP text encoder $\mathcal{E}_{\text{text}}^{\text{CLIP}}$ to maintain significant linguistic consistency and add an additional multilayer perceptron (MLP) layer $\Phi_t^{\text{adpt.}}$ as the feature adapter for pattern matching. With $\{\cdot\}_e$ denoting frozen weights, this process can be expressed as follows:

$$F_{\text{text}} = \Phi_t^{\text{adpt.}}\left(\left\{\mathcal{E}_{\text{text}}^{\text{CLIP}}\right\}_e(T_{\text{text}})\right) \tag{6}$$

where F_{text} denotes the task descriptors. With the task descriptors available, the next challenge is how to effectively incorporate this guidance into the HSI restoration process.

2) Frequency-Aware Feature Modulation: Based on the frequency domain analysis, as described in Section III-B, we design a feature modulation mechanism to mimic how degradations affect different frequency components. Given an input feature map F_f, we first decompose it into frequency components as follows:

$$F_f = F_f^l + F_f^h \tag{7}$$

where F_f^l and F_f^h denote the low- and high-frequency components, respectively, obtained through frequency-domain filtering.

Afterward, to merge text and image features, a previous study [66] proposed employing a single MLP layer, Ψ_m^i, to integrate them with intensity and bias embeddings. Consequently, our feature modulation can be represented as follows:

$$\lambda_m = \Psi_m^I(F_{\text{text}}),\ \mu_m = \Psi_m^{II}(F_{\text{text}}) \tag{8}$$

where Ψ_m^I and Ψ_m^{II} denote the chunk operations of Ψ_m to form the intensity and bias controller embeddings, respectively. To address the spectral consistency challenge in HSIs, our frequency-aware feature modulation adopts an affine transformation model [see (2)] tailored for spectral data, which is expressed as follows:

$$\hat{F}_f = \left(1 + \lambda_m^l\right) \odot F_f^l + \left(1 + \lambda_m^h\right) \odot F_f^h + \mu^l \tag{9}$$

where the intensity controller $\lambda \in [\lambda_m^l, \lambda_m^h]$ modulates the amplitude of high-frequency components to handle noise, and the Bias controller μ shifts the low-frequency baseline μ^l to compensate for blurring and band-missing effects, where $\mu = [\mu^l, \mu^h]$. This design ensures frequency-specific adjustments, which AiO RGB frameworks lack.

To validate the efficacy of this mechanism, we performed ablation studies demonstrating improved spectral fidelity and peak signal-to-noise ratio (PSNR) consistency across spectral bands, highlighting the need for spectral domain modulation to effectively manage complex HSI degradations.

C. Loss Function

Given the wide spectral range in HSIs, using only the ℓ_1 loss during training may not fully capture band variability, because it tends to prioritize bands with larger values, potentially compromising spectral integrity.

1) Spectral Angle Mapper Loss (SAM): The SAM metric is essential for material identification and is less sensitive to intensity variations. It emphasizes angular discrepancies over magnitudes

$$\ell_{\text{SAM}} = \frac{1}{N}\sum_{n=1}^{N}\cos^{-1}\left(\frac{I_n^T I_n^* + \epsilon}{\|I_n\|_2 \cdot \|I_n^*\|_2 + \epsilon}\right) \tag{10}$$

where I_n and I^*n are the nth spectral vectors of the ground-truth and reconstructed HSIs, respectively. ϵ is a small constant for numerical stability, and ℓ_{SAM} is computed in degrees.

2) Stationary Wavelet Transform Loss (SWT): SWT loss enforces frequency-domain consistency via wavelet sub-band alignment

$$\ell_{\text{SWT}} = \sum_j \lambda_j \left\|\text{SWT}(I)_j - \text{SWT}(I^*)_j\right\|_1 \tag{11}$$

with λ_j weighting the contribution of each sub-band.

3) BMSE Loss: The band-wise mean squared error (BMSE) loss balances reconstruction across all spectral bands to stabilize the optimization

$$\ell_{\text{BMSE}} = \frac{1}{N}\sum_{n=1}^{N}\|I_n - I_n^*\|_2. \tag{12}$$

4) Total Loss: The overall loss for training PromptHSI is defined as $\ell_{\text{total}} = \lambda_1\ell_1 + \lambda_2\ell_{\text{SAM}} + \lambda_3\ell_{\text{SWT}} + \lambda_4\ell_{\text{BMSE}}$, where the coefficients are set to $\lambda_1 = 1$, $\lambda_2 = 0.001$, $\lambda_3 = 0.01$, and $\lambda_4 = 0.01$ throughout all experiments.

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V. Degradation Synthesis of Dataset

To date, a comprehensive simulation framework for composite HSI degradation remains absent from the literature. Therefore, designing a realistic degradation process that accurately simulates real-world scenarios is critical. In this work, we adapt the gated degradation model (GDM) [71] from RGB image blind super-resolution to accommodate the unique characteristics of HSI restoration. The detailed configuration of our data generation pipeline is presented in Fig. 5.

A. Degradation Model

To simulate realistic hyperspectral imagery degradation, it is essential to consider the complete imaging acquisition pipeline. Specifically, observed images are initially affected by cloud coverage or shadow occlusion, resulting in obscured ground surfaces. Subsequently, hardware limitations alter the image resolution in both spatial and spectral domains. The imaging sensor then introduces random noise, and finally, atmospheric interference or bandwidth constraints during transmission manifest as stripe noise or band-missing artifacts.

1) Classical Hyperspectral Imaging Model: HSI restoration is an ill-posed problem where a clean image is corrupted by multiple degradation types. Formally, the degraded image I_{degraded} is generated from the clean image I_{clean} as

$$\begin{aligned} I_{\text{degraded}} &= D_{\text{stripe},n,s,c}\left(I_{\text{clean}}\right) \\ &= \left[\downarrow\left(c \oslash I_{\text{clean}}\right) + n\right] \odot M_{\text{stripe}} \end{aligned} \tag{13}$$

where $\oslash$ denotes the cloud occlusion blending operator, $\downarrow(\cdot)$ denotes the downsampling operator, n represents additive white Gaussian noise (AWGN), and $\odot$ denotes the masking operator applied by the stripe matrix M_{stripe}. In this process, the clean HSI I_{clean} is first degraded by cloud occlusion (c), then downsampled, further corrupted by noise n, and finally, some bands are randomly corrupted or missing due to stripe effects (M_{stripe}) during data transmission.

2) Probabilistic GDM: Following GDM [71], which synthesizes composite degradations for RGB images, we redesign the degradation types to better capture HSI characteristics. A gating mechanism is introduced to *randomly* select which degradation types are applied during synthesis. Formally, the probabilistic gated degradation process D_G is expressed as

$$\begin{aligned} I_{\text{degraded}} &= D_G\left(I_{\text{clean}}\right) \\ &= \left(\sigma_g\left(D_1\right) \circ \sigma_g\left(D_2\right) \circ \cdots \circ \sigma_g\left(D_m\right)\right)\left(I_{\text{clean}}\right) \end{aligned} \tag{14}$$

where

$$\sigma_g\left(D_i\right)\left(r^d\right) = \begin{cases} D_i\left(r^d\right), & \text{if } \ g_i \sim \text{Bernoulli}\left(p\right) \\ r^d, & \text{if } \ g_i = 0 \end{cases} \tag{15}$$

and r^d denotes the input HSI. Each gate g_i is activated with probability p (e.g., $p = 0.5$), determining whether degradation type D_i is applied. Probabilistic GDM (PGDM) enables diverse integration of fundamental degradation types, making the synthesized HSI more realistic for real-world scenarios.

B. Degradation Simulation

This section presents a detailed discussion of the considerations and procedures involved in modeling each specific degradation type. We begin by providing the necessary background and subsequently describe the synthesis methodology for each degradation.

Collecting large-scale real-world paired HSI datasets with ground truth is extremely challenging due to several factors. First, satellite revisit cycles and temporal variations between acquisitions make it difficult to obtain perfectly aligned clean-degraded pairs. Second, atmospheric conditions, sensor calibration drift, and scene dynamics introduce inconsistencies that compromise pair quality. Third, most operational satellites do not provide simultaneous clean reference data alongside degraded observations. These practical constraints motivated our simulation-based approach using PGDM, which enables controlled generation of diverse composite degradations while maintaining spectral–spatial fidelity for training and evaluation.

1) Cloud Occlusion:

1) *Background:* Optical satellite imagery is widely used across various applications but is often hindered by adverse weather conditions, particularly cloud cover, which obscures surface features of interest. Assembling high-quality cloud-free datasets remains challenging, motivating the need for synthetic cloudy data generation for reliable evaluation.
2) *Synthesis:* Following the cloud synthesis methodology introduced in the *Satellite Cloud Generator*[1] [11], we employ a pair of real HSIs: one containing clouds and another cloud-free neighboring image. The cloud mask detector proposed in [56] is applied to estimate cloud-covered regions in Sentinel-2 spectral bands. This mask is used to approximate the cloud endmember function across different spectra. Finally, synthetic clouds of varying intensities (thin or thick) are generated on the clean HSI, as illustrated in Fig. 6. The cloudy HSI generation process is expressed as

$$I_{\text{cloudy}} = \left(1 - M_C\right) \cdot I_{\text{clean}} + M_C \cdot I_{\text{cloud}} \tag{16}$$

where M_C denotes a spectrally varying mask characterizing the fractional cloud layer contribution at each pixel and wavelength, and I_{cloud} denotes the cloud component synthesized using Perlin noise and adjusted to approximate the spectral endmember of clouds across different bands.

2) Spatial and Spectral Blurring:

1) *Background:* Due to diverse sensor payloads and varying orbital altitudes of LEO satellites, remote sensing data exhibits significant heterogeneity in both spatial and spectral resolutions. Hyperspectral imaging sensors typically suffer from limited spatial resolution due to technical and cost constraints, hindering detailed surface feature observation. To address these limitations, we incorporate both spatial and spectral super-resolution tasks into our model.

[1]Official repository: https://github.com/strath-ai/SatelliteCloudGenerator

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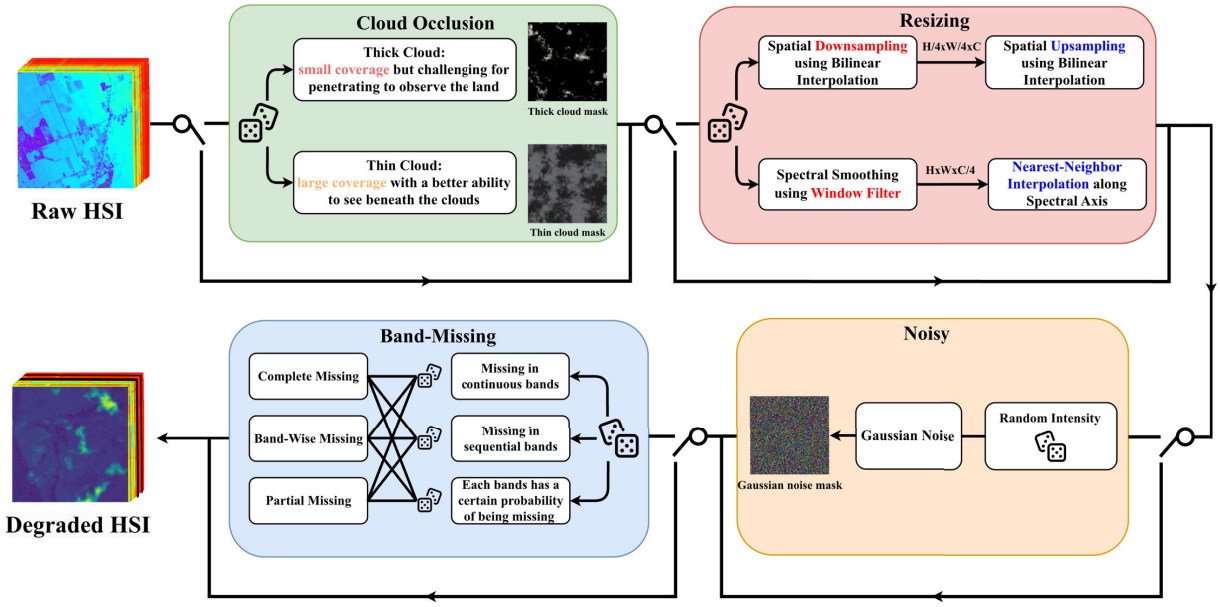


Fig. 5. Proposed dataset synthesis pipeline. Each degradation is activated with probability p (set to 0.5 for all experiments). When activated, each degradation type randomly selects one of its subdegradations to apply to the input HSI, ensuring diverse synthetic data generation.

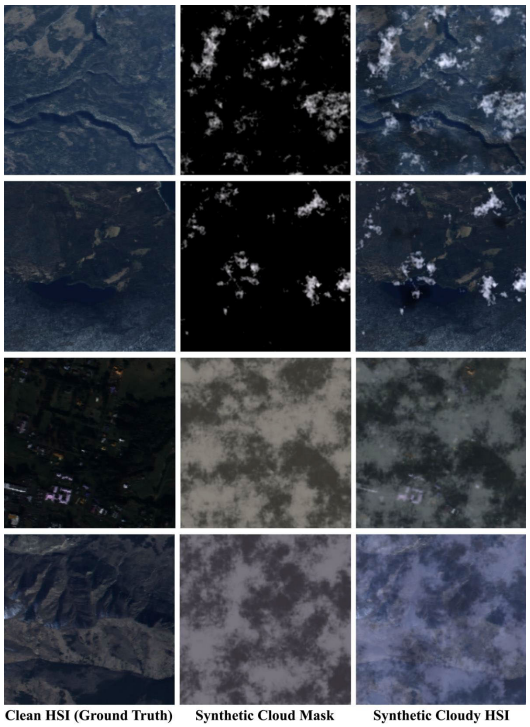


Fig. 6. Visualization of synthetic cloudy HSI. The top two rows show *thick* cloud cover, which obscures a small area and makes underlying features difficult to observe. The bottom two rows show *thin* clouds, which cover a larger area but allow some surface details to remain visible, creating a fog-like effect.

2) *Synthesis:* To simulate resolution degradation, we implement a downsampling–upsampling procedure across spatial and spectral domains. The spatial blurring process is defined as

$$I_{\text{spatial blur}} = I_{\text{clean}}\text{Spa}_{\downarrow}^{4}\text{Spa}_{\uparrow}^{4} \tag{17}$$

where $\text{Spa}_{\downarrow}$ and $\text{Spa}_{\uparrow}$ denote spatial bilinear downsampling and upsampling with a factor of 4, respectively. The spectral blurring process is expressed as

$$I_{\text{spectral blur}} = (\Phi(I_{\text{clean}}, 5, 4))_{\text{Spe}_{\uparrow}^{4}} \tag{18}$$

where $\Phi(\cdot, \omega, \varsigma)$ denotes a sliding Gaussian window filter with window size ω and stride ς to reduce spectral resolution, and $\text{Spe}_{\uparrow}$ denotes nearest-neighbor interpolation with a factor of 4 along the spectral dimension to restore the original spectral size.

3) Random Noise:

1) *Background:* HSIs typically contain noise due to environmental interference, sensor quality limitations, or signal transmission issues. Therefore, HSI denoising is crucial for HSI restoration and downstream applications.
2) *Synthesis:* To enhance the diversity of noise magnitudes in our dataset, we simulate AWGN using signal-to-noise ratio (SNR) values sampled from a normal distribution with mean μ and standard deviation σ. The Gaussian noise I_ϵ, randomly drawn from $\mathcal{N}(\mathbf{0}, \mathbf{1})$, is scaled by the intensity term to generate AWGN. The noisy HSI is formulated as

$$I_{\text{noisy}} = I_{\text{clean}} + N_{\text{awgn}} = I_{\text{clean}} + I_\epsilon \cdot \sqrt{\frac{\|I_{\text{clean}}\|_2^2}{\text{SNR}}}. \tag{19}$$

4) Band-Missing Degradation:

1) *Background:* Beyond random noise, HSIs are frequently affected by band-missing or stripe-like degradations arising from sensor malfunctions or transmission interference [69]. Such artifacts typically appear as missing spectral bands or stripes aligned with the sensor scanning direction, disrupting spectral continuity and impairing subsequent analysis tasks.
2) *Synthesis:* We simulate this degradation and categorize it into three types [19]: 1) band complete missing, where pixel values of entire bands are completely missing; 2) band-wise missing, where pixel values in certain rows

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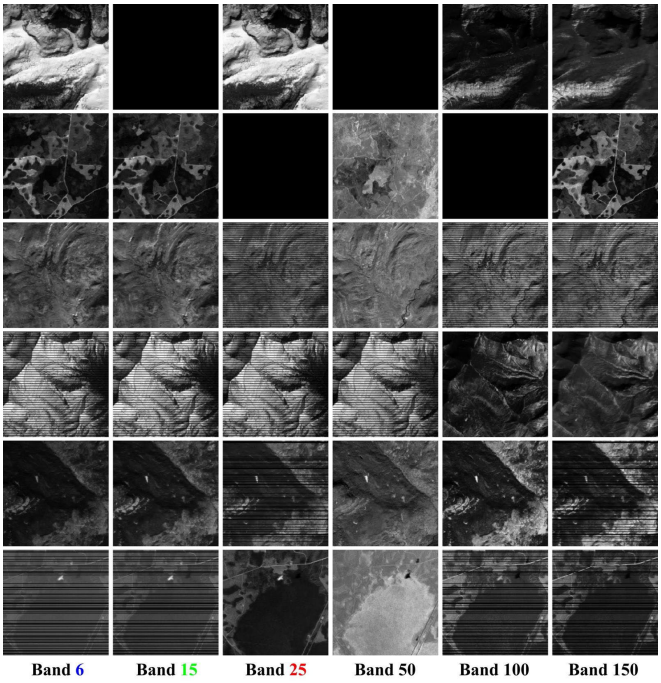


Fig. 7. Visualization of simulated band-missing on HSI. We selected multiple bands to demonstrate different types of band-missing degradations, with the 6th, 15th, and 25th bands representing B-, G-, and R-bands, respectively. The first two rows depict band completely missing, the next two rows represent band-wise missing, and the final two rows represent partial missing.

disappear in a regular pattern; and 3) partial missing, where rows disappear in an irregular pattern, creating stripe-like defects. The visualization of these three missing types is shown in Fig. 7.

VI. Experimental Results

A. Experiment Setup

1) Data Description: We conducted our experiments using the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) dataset,[2] provided by [19] and [21]. This is a large-scale HSI dataset that includes numerous spectral bands across diverse geographical areas. The dataset contains 2078 HSIs, which were randomly split into training (1678 images), validation (200 images), and testing (200 images) sets. The original HSIs were partitioned into distinct subimages, each measuring 259 × 259 pixels, comprising 224 spectral bands spanning wavelengths from 400 to 2500 nm. Consistent with previous research [19], [21], spectral bands identified as lower quality (1–10, 104–116, 152–170, and 215–224) were excluded, resulting in HSIs with 172 spectral bands.

Following the data generation pipeline described in Section V, all clean HSIs were preprocessed to synthesize the composite degraded HSIs, along with the corresponding text descriptions, as shown in Table I. For all experiments, we used the short prompt format for text descriptions. The generated dataset, named *HyperAD*,[3] is publicly available to facilitate further research.

[2]AVIRIS Data Portal:https://aviris.jpl.nasa.gov/data/index.html
[3]HyperAD: https://github.com/chingheng0808/HyperAD

TABLE I

Text Prompt Formats. Different Colors Represent Various Degradation Types. If a Particular Degradation Type Is Absent, It Is Excluded From the Text Prompt. The Notation [] Specifies the Subtype of the Respective Degradation, and N_{MB} Denotes the Number of Bands Affected by Band-Missing Degradation

Prompt type	Prompt format
Long prompt	"This hyperspectral image faces '[*band-missing type*]' on N_{mb} bands; it also confronts '[*cloudy type*]', '[*noisy*]'; besides, there exists 'blurring effect in [*blurring type*] domain'."
Short prompt	"[*cloudy type*], [*noisy*], [*blurring type*], [*band-missing type*]"

2) Hyperparameter Setting: We used a single NVIDIA-RTX3090 GPU for training. The batch size was set to 8, and the number of training epochs was fixed at 300 for all experiments involving the proposed method. The AdamW optimizer [42] ($\beta_1 = 0.9$, $\beta_2 = 0.999$, weight decay 1×10^{-2}) was employed with an initial learning rate of 3×10^{-4}. The learning rate was dynamically adjusted during the training process using a step-based learning rate scheduler. The scheduler reduced the learning rate by a factor of 0.5 every 50 epochs to ensure stable convergence.

3) Data Augmentation: The input HSI was randomly cropped to patches with a size of $112 \times 112 \times 172$, incorporating random horizontal and vertical rotations, and flipping during the training phase. During inference, the model was evaluated using the input dimensions of $224 \times 224 \times 172$.

B. Quantitative Results and Analysis

To evaluate the performance of PromptHSI, we compared it with ten supervised HSI restoration methods for specific degradation types, including HSIDwRD [70], SERT [32], KSSANet [27], SRDNet [41], SNLSR [22], SGNet [44], AACNet [64], SST [31], RTCS [19], and HSDT [26] on the synthesized AVIRIS dataset with composite degradations. Because there are no existing AiO HSI restoration methods, we compared our approach with AiO RGB image restoration methods in the experiments, including AirNet [28], PromptIR [48], HAIR [4], InstructIR [9], and OneRestore [15].

To ensure a fair comparison with the hyperspectral data, the input and output channel dimensions of these RGB-domain methods were expanded to match the number of spectral bands used in our experiments. The model performance was evaluated using four metrics: PSNR, SAM, root-mean-square error (RMSE), and Erreur Relative Globale Adimensionnelle de Synthèse (ERGAS). The quantitative results are summarized in Table II. Fig. 8 qualitatively compares the proposed method with task-specific HSI restoration approaches. Fig. 9 compares the proposed method with AiO RGB restoration methods.

1) Ours Versus Task-Specific HSI Restoration Methods: All comparison methods cannot effectively recover the degraded HSIs because of their inability to handle different or unseen degradations. In contrast, PromptHSI achieves better restoration even without text prompts (Table III), demonstrating the capability of our unified restoration network design.

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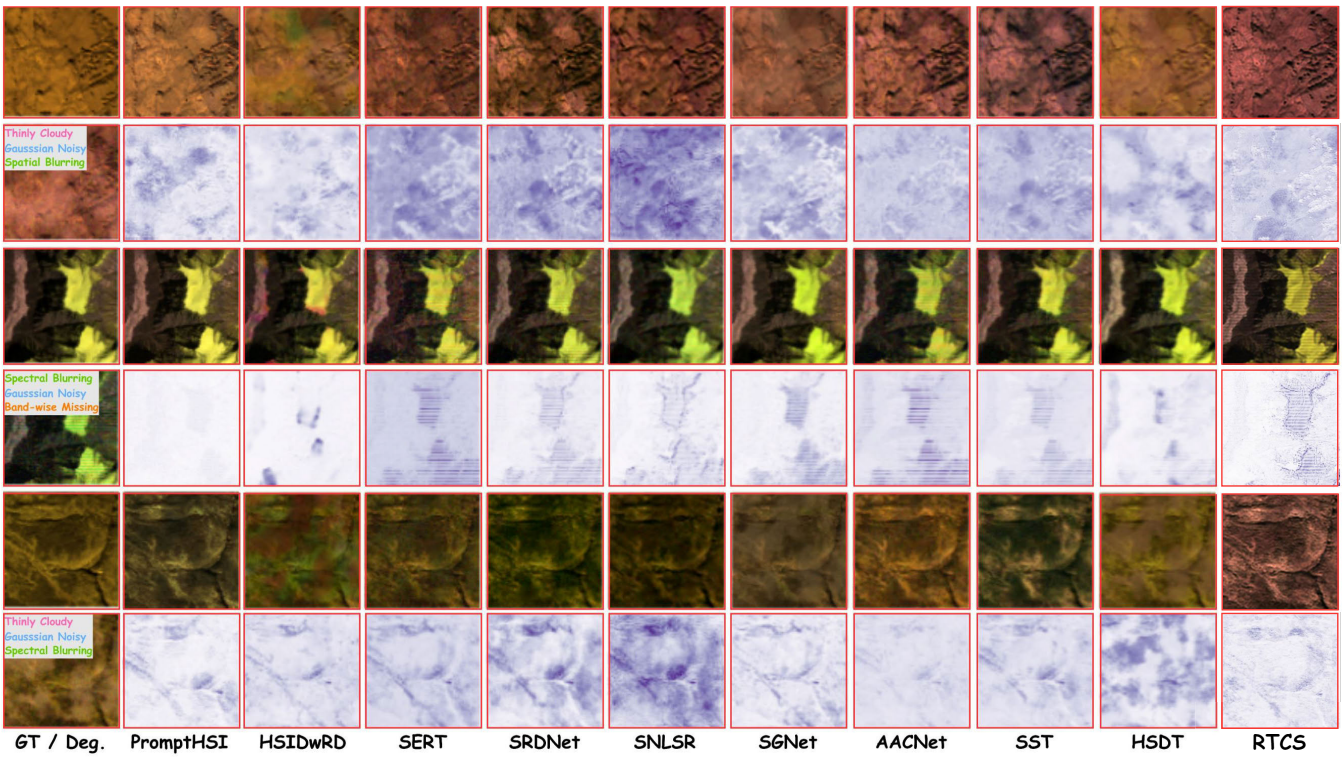


Fig. 8. Visualization results for HSI restoration methods. (Top) Reconstructed HSI and (bottom) residual image derived from ground-truth and reconstructed result. Most of the state-of-the-art HSI restoration methods fail to deal with composite degradations.

TABLE II

OVERALL PERFORMANCE EVALUATION FOR COMPOSITE DEGRADATION AND COMPLEXITY COMPARISON.TO SIMULATE COMPOSITE DEGRADATIONS, THE p VALUE WAS FIXED AT 0.5 FOR TRIGGERING INDIVIDUAL DEGRADATION TYPES (FIG. 5), FOR ALL EXPERIMENTS. NOTE THAT **CR.** INDICATES WHETHER THE METHOD IS **CONTROLLABLE**. AS FOR THE COMPLEXITY PARTS, M AND G INDICATE 10^6 AND 10^9, RESPECTIVELY. THE BEST, SECOND-BEST, AND THIRD-BEST RESULTS ARE HIGHLIGHTED IN **BOLD-RED**, **BOLD-BLUE**, AND **BOLD-GREEN**, RESPECTIVELY

Type	Methods	Venue & Year	Task	Domain	Cr.	PSNR↑	SAM↓	RMSE↓	ERGAS↓	#Params↓	FLOPs↓
Input	-	-	-	-	×	15.2454	17.7463	0.0643	33.7220	-	-
One-to-One	HSIDwRD [70]	ICCV 2021	Denoising	HSI	×	20.6270	12.7891	0.0364	97.3601	23.62M	5054.16G
	SERT [32]	CVPR 2023	Denoising	HSI	×	16.1657	15.9853	0.0348	29.2053	2.74M	176.06 G
	SRDNet [41]	TGRS 2024	Super-Resolution	HSI	×	19.0520	11.9590	0.0320	**16.5861**	1.86M	3771.80G
	SNLSR [22]	TIP 2024	Super-Resolution	HSI	×	17.8785	20.4389	0.0445	19.1267	1.75M	14.20G
	KSSANet [27]	ICASSP 2025	Super-Resolution	HSI	×	15.6778	17.1524	0.0393	33.2496	5.03M	252.02G
	SGNet [44]	ISPRS 2022	Decloud	HSI	×	18.0694	12.9476	0.0359	17.9773	0.54M	57.39G
	AACNet [64]	TGRS 2023	Decloud	HSI	×	17.6279	14.5962	0.0324	26.2115	2.76M	171.23G
	SST [31]	AAAI 2023	Inpainting	HSI	×	18.9792	10.9860	**0.0244**	26.9934	4.30M	281.72G
	HSDT [26]	ICCV 2023	Inpainting	HSI	×	18.0665	11.3451	0.0372	68.4451	0.51M	67.55G
	RTCS [19]	TGRS 2024	Inpainting/Denoising	HSI	×	19.9596	8.9775	0.0288	20.8981	7.24M	36.03G
AiO for specific	AirNet [28]	CVPR 2022	-	RGB	×	**23.6647**	**7.5970**	**0.0233**	**16.4194**	6.07M	245.86G
	PromptIR [48]	NeurIPS 2023	-	RGB	×	17.2667	**9.6437**	0.0405	29.2768	33.18M	132.07G
	HAIR [4]	arXiv 2024	-	RGB	×	**20.5779**	14.4329	0.0275	24.4211	7.86M	42.51G
	InstructIR [9]	ECCV 2024	-	RGB	✓	20.3106	14.5727	0.0306	27.5454	16.02M	17.28G
AiO for composite	OneRestore [15]	ECCV 2024	-	RGB	✓	17.3321	15.0320	0.0271	52.1162	5.85M	12.05G
	PromptHSI (Ours)	-	-	HSI	✓	**26.4432**	**6.0601**	**0.0185**	**8.3566**	26.14M	135.83G

TABLE III

ABLATION STUDIES ON NETWORK ARCHITECTURE COMPONENTS

Without	PSNR↑	SAM↓	RMSE↓	ERGAS↓
PGFM	25.49	6.63	0.0205	**13.21**
Adapter	**25.87**	**6.36**	**0.0188**	14.36
Enhancement	23.75	7.99	0.0262	16.00
Full model	**26.44**	**6.06**	**0.0185**	**8.36**

2) Ours Versus AiO RGB Restoration Methods: Unlike AiO RGB image restoration methods that rely on uniform feature modulation, the frequency-aware modulation of our PromptHSI framework dynamically tailors the restoration to each spectral band, thereby achieving cross-band consistency, which is crucial for HSI applications. In general, our PromptHSI outperforms the comparison AiO methods. First, most of the existing methods struggle to handle composite degradations. As previously discussed, both the data and features extracted by pretrained CLIP [50] exhibit domain gaps with HSI. Therefore, these models are not effective in AiO HSI restoration. A crucial observation lies in the failure of OneRestore [15]. Its strategy of injecting task descriptions directly into the spatial encoder disrupted the delicate spatial–spectral correlations inherent in HSI, leading to significant artifacts (decoding failure). This failure mode highlights the necessity of our decoupled approach: by modulating features in the

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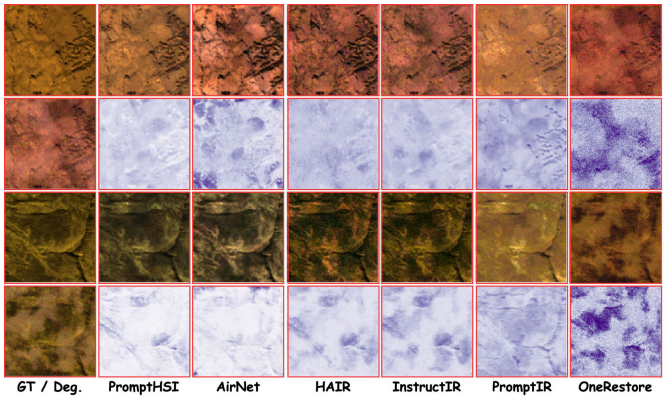


Fig. 9. Comparison with AiO RGB image restoration methods. For the text-prompt-based methods [9], [15] and the PromptHSI method, we employed the standard text-prompts presented in this study for network training and inference for related experiments.

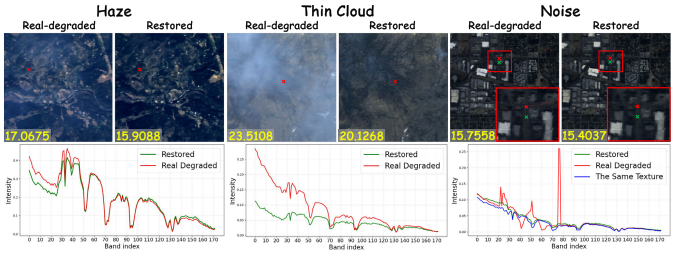


Fig. 10. No-reference evaluation on real-world AVIRIS scenes. The red × markers show locations of plotted spectra. The green × indicates a reference road region. Per-image quality scores (lower is better) are annotated. PromptHSI effectively removes haze/cloud and stripe noise while preserving spectral signatures.

frequency domain rather than the spatial domain, PromptHSI effectively bypasses this interference, maintaining spectral integrity while adapting to diverse degradations.

C. Evaluation on Real-World Data

While quantitative metrics on synthetic data provide objective baselines, real-world sensing environments introduce complex, unmodeled artifacts. To validate PromptHSI's robustness beyond the simulation envelope, we conducted qualitative evaluations on real airborne AVIRIS scenes where ground truth is unavailable. We adopt the no-reference HSI quality metric of Yang et al. [65], which encodes spectral (generalized Gaussian distribution (GGD)-fit statistics of locally normalized spectra) and spatial cues (multiscale Log-Gabor responses and gradients) as a multivariate Gaussian, and scores quality via a modified Bhattacharyya distance to a pristine reference model. Lower scores indicate better quality. Unlike RGB-oriented no-reference metrics that assess only spatial artifacts, this HSI-specific measure jointly evaluates spatial fidelity and spectral consistency.

Fig. 10 shows representative results on severely degraded real scenes. The red × markers indicate spectral curve locations. PromptHSI effectively attenuates haze and cloud cover while suppressing structured stripe noise—artifacts characteristic of pushbroom sensors that differ from synthetic training data. The per-image no-reference scores annotated in each panel corroborate these visual improvements. The green × marker indicates a reference road region where restored spectra

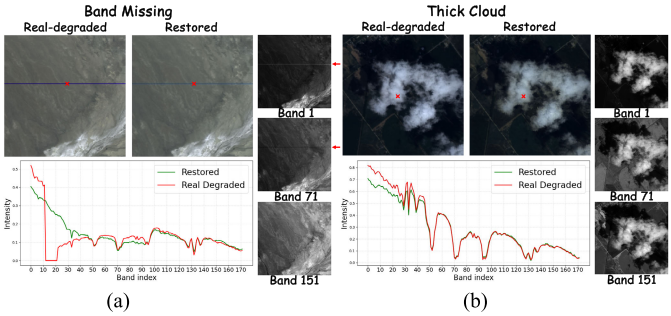


Fig. 11. OOD challenges. Failure cases on real HSI samples from the dataset. Two representative OOD scenarios where our method struggles. (a) Abnormal band-missing patterns caused by sensor malfunctions. (b) Extreme cloud-dominated scenes that exceed the typical training range.

remain consistent with expected asphalt signatures, validating spectral fidelity preservation. These results demonstrate robust generalization despite the domain gap between PGDM-synthesized training data and real sensor artifacts.

1) OOD Challenges: However, in some cases, PromptHSI encounters difficulties under extreme out-of-distribution (OOD) conditions, which is a fundamental challenge stemming from HSI data scarcity and the vast diversity of real-world degradation patterns across sensors, atmospheric conditions, and scene types. Fig. 11 illustrates two representative OOD failure modes: 1) abnormal band-missing patterns where real sensor malfunctions produce nonzero structured artifacts (e.g., stripes in bands 1 and 81) rather than zero-valued patterns in simulation; and 2) extreme cloud-dominated scenes where opacity far exceeds typical training ranges, preventing reliable signal recovery. These cases highlight the inherent difficulty of modeling long-tailed real-world degradations. Indicating that addressing these challenges will require future advances in either comprehensive degradation modeling informed by real sensor telemetry or robust architectures with stronger physical inductive biases.

D. Ablation Studies

The experimental results demonstrate the effectiveness of the proposed PromptHSI framework under composite degradations. We further conducted ablation studies to investigate the impact of different components and strategies.

1) Network Architecture Components:

1) *Ablation of Network Components:* To understand the impact of the network components, we conducted ablation studies by removing PGFM, adapter, and spatial–spectral enhancement components from the network. The results are presented in Table III. Even without textual features, the proposed PromptHSI framework outperforms peer methods in handling composite degradation, demonstrating the effectiveness and universality of its network architecture. The adapter proves crucial in bridging the domain gap between VLM and HSI features via pattern matching, thereby improving global reconstruction performance. This improvement is particularly evident in the ERGAS metric, which shows a significant gain of 6.0 points when the adapter is

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TABLE IV
ABLATION STUDIES ON FREQUENCY MODULATION STRATEGY

Strategy	PSNR↑	SAM↓	RMSE↓	ERGAS↓
Intensity-only	**26.01**	**6.36**	**0.0187**	**9.76**
Bias-only	25.64	6.40	0.0190	12.50
Weighted-sum	25.73	6.36	0.0188	14.36
Intensity & Bias	**26.44**	**6.06**	**0.0185**	**8.36**

TABLE V
ABLATION STUDIES ON PROMPT MODALITY

Type	Cr.	PSNR↑	SAM↓	RMSE↓	ERGAS↓
RGB to VLM	×	24.13	7.62	0.0257	14.52
Hybrid	✓	**25.24**	**6.90**	**0.0191**	**11.44**
PGB&PIM [48]	×	23.06	9.49	0.0287	24.56
Text to VLM	✓	**26.44**	**6.06**	**0.0185**	**8.36**

included. In addition, a performance decline is observed when spatial–spectral enhancement is removed, indicating that feature enhancement is essential for detail recovery within the cascade decoder. More detailed ablation studies of the model components are presented in the Supplementary Material.

2) *Integration Strategy of Frequency-Aware Modulation:* We investigated whether independent control of frequency components yields better adaptation. Table IV reveals that modulating both intensity (λ) and bias (μ) concurrently achieves the optimal tradeoff (PSNR 26.44 dB). Interestingly, the *Intensity-only* configuration contributes more significantly to performance than *Bias-only*, suggesting that amplitude adjustment plays a dominant role in suppressing noise and blurring artifacts.

2) Prompt Design and Controllability:

1) *Visual- or Text-Prompts:* The prompting type is crucial for recognizing degradations and guiding model behavior. In our first configuration, we use RGB bands from HSIs as visual prompts, encoded via CLIP [50] and guided by PGFM. In the second configuration, we combine RGB and textual prompts to form hybrid task descriptors. In the final configuration, PGFM is removed, and we employ the PIM and PGM blocks from PromptIR [48] to generate visual prompts and feature interaction. Table V shows that composite degradations cause significant information loss, making it difficult to obtain compact task descriptors or effective visual prompts. Incorporating textual features overcomes these challenges by providing robust semantic guidance, achieving substantial improvements.
2) *Effect of Text-Prompt Length:* We further evaluated the impact of long versus short text prompts on model performance, as shown in Table VI. The short prompt format (see Table I) achieves better performance than the long prompt. While the long text prompt offers distinct descriptions for each degradation type, combining them into a single statement that more closely resembles natural human language, the short prompt's concise format proves more effective for guiding the restoration process.
3) *Controllable Prompts:* Fig. 12 shows the restoration results for a composite of three degradation types using the corresponding prompt combinations. The results indicate that PromptHSI is capable of a certain degree of restoration even without any prompts, as evidenced by the recovery of missing channels when no prompt is provided. However, when the cloudy prompt is omitted, residual degradation remains more obvious, indicating that cloud-related degradation is the most dominant among the three degradation types. Table VI shows that prompts significantly affect restoration performance. We evaluate two stress settings—no prompt (no text provided) and random prompt (a randomly generated prompt per HSI). Both lead to a pronounced drop across all metrics, indicating that prompts are both important and controllable in PromptHSI.

TABLE VI
ABLATION STUDIES ON PROMPT CONFIGURATION

Configuration	PSNR↑	SAM↓	RMSE↓	ERGAS↓
Prompt length (corresponding to Tables I)				
Long	**25.88**	**6.68**	**0.0196**	**10.79**
Short	**26.44**	**6.06**	**0.0186**	**8.36**
Prompt availability				
None	18.96	9.54	0.0505	34.91
Random	**19.27**	**9.66**	**0.0426**	**28.58**
Correct	**26.44**	**6.06**	**0.0185**	**8.36**

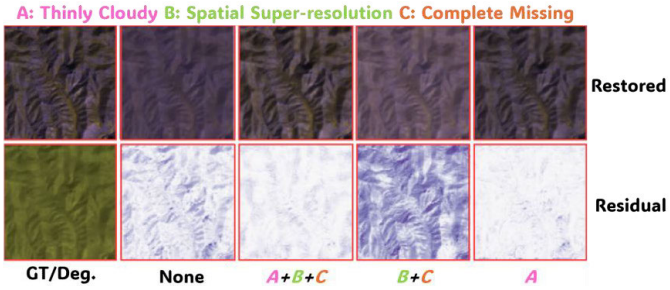


Fig. 12. Visualization results for controllability. In the case of composite degradation, (Top) row shows HSIs restored using specific prompts, with the first column representing the ground truth and (bottom) row presents the corresponding residual maps, which illustrate the differences between the reconstructed HSIs and the ground truth, while the first column displays the composite degradation HSI.

E. Analysis and Robustness

1) Degradation Discriminability:

1) *t-SNE Visualization:* Does PromptHSI truly understand the degradation semantics? To answer this, we visualize the learned feature space using t-SNE, a standard technique in AiO methods [4], [9], [10], [48]. Fig. 13 shows the t-SNE embeddings for both multiple and composite degradations, demonstrating superior discriminative ability with well-separated degradation clusters. Additional comparisons are provided in the

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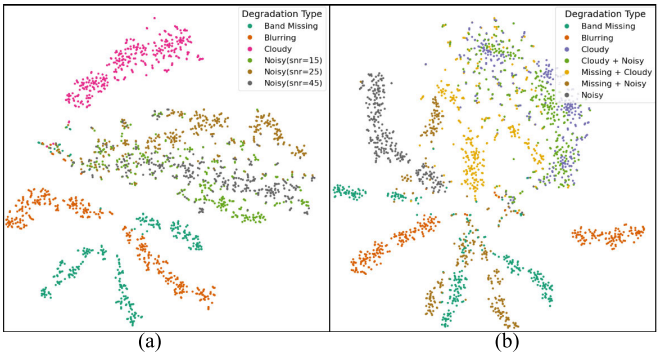


Fig. 13. Comparison of t-SNE visualization results for the degradation embeddings of the proposed PromptHSI trained on individual degradation types (a) multiple and (b) composite degradations. The proposed PromptHSI method demonstrated its discriminative ability in identifying different degradation types, including unseen degradation combinations, during the testing phase.

TABLE VII

QUANTITATIVE CLUSTER SEPARABILITY OF MODELS' FEATURE EMBEDDINGS COMPARED WITH AIO METHODS

Metric	**PromptHSI**	AirNet	PromptIR	HAIR	InstructIR
Silhouette↑	**0.4868**	-0.0786	-0.0248	-0.0163	**0.0098**
NMI↑	**0.7026**	0.1737	0.0849	0.2027	**0.4225**

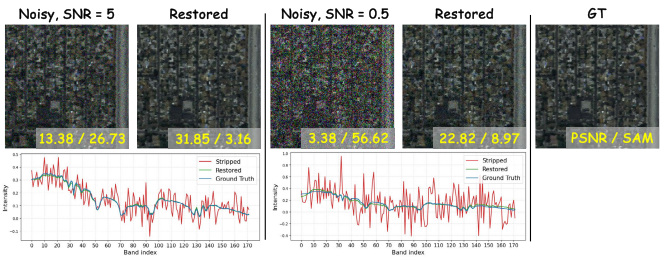


Fig. 14. Stress-test visualizations under severe noise. Severity is controlled by the SNR defined in (19). Spectral signatures display the central pixel in each HSI.

Supplementary Material. These results confirm that our text prompts modality can effectively distinguish degradation types compared to conventional prompt-based AiO RGB methods.

2) *Quantitative Cluster Separability:* We further assess degradation type separability using the Silhouette score [52] and NMI [46], computed on each model's feature embeddings across multiple degradation types. Table VII shows that most methods fail to clearly separate different degradations, whereas PromptHSI achieves the highest separability, benefiting from text-prompt guidance.

2) Stress Tests on Severe Degradations: We stress-test *PromptHSI* under two OOD conditions that frequently occur in practice.

1) *Severe Noise:* We decrease the SNR in 19 to synthesize increasingly heavy sensor noise; qualitative results are shown in Fig. 14. As SNR decreases, PromptHSI degrades gracefully: structural details remain discernible, and residual energy concentrates in high-frequency regions rather than causing band-wise drift.

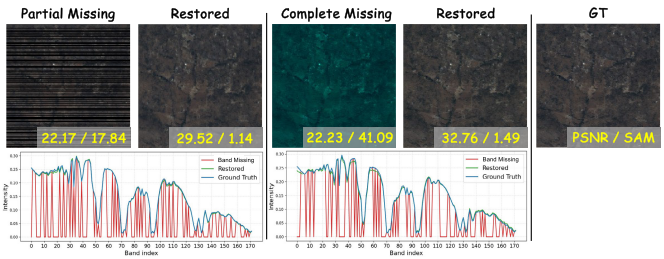


Fig. 15. Stress-test visualizations under severe band-missing. We simulate severe partial/complete band-missing by using higher per-band drop probabilities. Spectral signatures display the central pixel in each HSI.

2) *Severe Band-Missing:* We increase the per-band drop probability to induce partial and complete spectral holes. Visualizations are shown in Fig. 15. PromptHSI preserves cross-band consistency and plausibly inpaints the missing responses. The reflectance plots provide supporting evidence, where the restored spectra fill the gaps and closely match the ground truth.

Overall, PromptHSI restores HSIs effectively even under severe degradations and shows strong generalization. These results demonstrate the superior capability of PromptHSI in maintaining spectral fidelity across multiple degradation types, highlighting the necessity of frequency-aware modulation, particularly when compared to AiO RGB methods that exhibit significant performance degradation in the HSI domain. In summary, PromptHSI offers a universal and controllable solution for addressing composite degradation in real-world applications. Additional visualizations and further analyses are presented in the Supplementary Material.

VII. FUTURE WORK

While PromptHSI demonstrates strong performance on diverse degradation types, several directions warrant further investigation to enhance controllability, real-world generalization, and computational efficiency.

A. Enhanced Controllability

Although our method effectively separates degradation classes via text prompts, empty prompts can still trigger minor restoration actions rather than strict identity mappings. Future work will explore contrastive learning objectives and explicit no-op regularization to enforce stricter alignment between prompt semantics and restoration behavior, ensuring that null prompts produce true no-operation outputs.

B. Improved Real-World Generalization

As demonstrated in Section VI-C, extreme OOD degradations—particularly abnormal sensor artifacts and severe atmospheric conditions—pose challenges for simulation-based training. Addressing this requires two complementary strategies: 1) enriching PGDM with more comprehensive degradation models informed by real sensor telemetry and failure mode analysis; and 2) incorporating physics-aware priors (e.g., atmospheric radiative transfer

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models, sensor point spread functions) combined with unsupervised domain adaptation techniques to bridge the gap between synthetic training data and real-world sensor characteristics.

C. Computational Efficiency

The transformer-based architecture in PGFAB incurs substantial computational costs, limiting deployment on resource-constrained platforms. Future research will investigate model compression strategies, including knowledge distillation, structured pruning, and quantization-aware training. In addition, we will explore efficient alternatives such as state-space models or hybrid CNN-transformer architectures to achieve favorable accuracy-efficiency tradeoffs for practical applications.

VIII. Conclusion

This work presents PromptHSI, a universal framework for HSI restoration addressing composite degradations through text-guided, frequency-aware modulation. Unlike existing AiO RGB restoration methods, PromptHSI is specifically designed to leverage the spectral–spatial structure of hyperspectral data. By integrating VLM-guided prompt learning with prompt-guided frequency modulation, our method enables adaptive and controllable restoration across diverse degradation scenarios.

Extensive experiments demonstrate that PromptHSI outperforms state-of-the-art methods across multiple degradation types—cloud occlusion, noise, blurring, and band-missing—while maintaining performance under composite degradations. Evaluation on real AVIRIS scenes demonstrates practical applicability. To facilitate future research, we contribute a comprehensive benchmark dataset for composite HSI degradation using our synthetic dataset, which will be publicly released. PromptHSI provides a foundation for controllable hyperspectral restoration and demonstrates the potential for applying vision–language models to remote sensing applications, paving the way for future research in this field.

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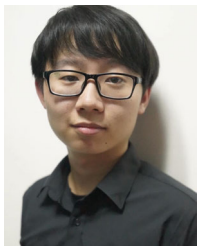
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